TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

High Resistance by the Method of Deflection

When resistance climbs toward a megohm, bridges starve: the currents shrink below what balance methods can sense. A sensitive galvanometer, calibrated in amperes per division, reads the tiny current directly — and Ohm's law does the rest.

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1. Aim and theory

Aim: to measure a high resistance (of order 10^5 Ω) by the deflection method. A galvanometer whose **figure of merit** k (current per scale division) and resistance G are already known — Experiment 1's half-deflection work — is connected in series with the unknown high resistance X and a battery of known emf E. The steady deflection θ gives the current directly:

$$I = k\,\theta$$

and the circuit's total resistance follows from Ohm's law, $E = I(X + G + r)$ with r the battery's small internal resistance:

$$X = E/(k\,\theta) - G - r \approx E/(k\,\theta) - G$$

The method succeeds precisely where null methods fail: it needs only a measurable deflection, not a balance, so the smaller the current the method still works — provided the galvanometer is sensitive enough (k of order 10^{-6} A/div or less) and stray leakage paths do not offer the current an easier route than X itself.

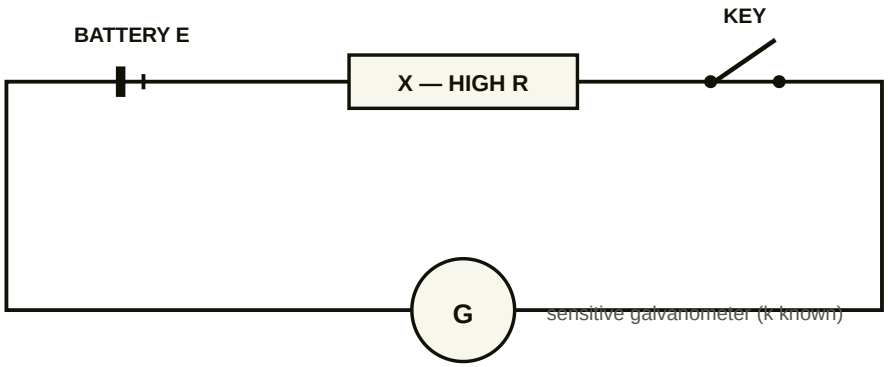


Figure 1 — The whole instrument: battery, key, the high resistance and the calibrated galvanometer in one series loop. Nothing balances; the needle simply reports the current.

2. Apparatus and procedure

- **Apparatus:** the high resistance to be measured (e.g. a 0.1 MΩ resistor or a length of high-resistivity wire), sensitive galvanometer with known k and G, battery of several cells (E known), key, well-insulated connecting leads.
- Determine k for the galvanometer beforehand (half-deflection against a known resistance, Experiment 1) — or take the calibrated value for the instrument in use.
- Connect battery, key, X and galvanometer in series (Fig. 1), keeping leads short, dry and off the bench surface where possible (leakage!).
- Close the key; read the steady deflection θ (both needle ends; reverse the current with a commutator and average to cancel zero error).
- Repeat with a second battery voltage if available; compute X from X = E/(kθ) − G each time and average.
- Sanity check: repeat with a known standard high resistance in place of X and confirm the method recovers it.

3. Worked calculation

Galvanometer: k = 2.0×10^{-6} A/div, G = 100 Ω. Battery E = 6.0 V. Mean deflection θ = 25 divisions.

$$I = k\theta = 2.0 \times 10^{-6} \times 25 = 5.0 \times 10^{-5}\ \text{A}$$

$$X = E/I - G = 6.0/(5.0 \times 10^{-5}) - 100 = 120{,}000 - 100 = 119{,}900\ \Omega \approx 1.20 \times 10^{5}\ \Omega$$

Note how G, a full hundred ohms, barely dents the answer (0.08 %) — at high resistances the galvanometer's own resistance stops mattering, which is part of the method's charm.

4. Precautions

- Leakage is the silent error: damp bench tops, frayed insulation and finger-prints across terminals all offer paths in parallel with X, and at 10^5 Ω even a 10^7 Ω damp path steals 1 %. Work dry, handle insulation only, keep leads airborne on insulating stands for the critical stretch.
- The emf must be known and steady — X is computed from E directly, so a tired battery falsifies everything.
- Let the deflection settle fully; a sensitive galvanometer is slow, and reading early understates θ (overstates X).
- Reverse and average to remove zero error; shield the galvanometer from draughts and vibration.

Practice (viva preparation)

1. Why does a Wheatstone bridge become impractical at 10^5–10^6 Ω while this method still works?
2. With k = 1.5 × 10^{-6} A/div, G = 80 Ω, E = 4 V and θ = 20 div, find X.
3. A leakage path of 2 × 10^6 Ω appears across a true X of 1.2 × 10^5 Ω. What value does the experiment report?
4. Why does G matter so little here when it mattered so much in Experiment 1?
5. How would you adapt the method for an even higher resistance, say 10^7 Ω?

Answer key

1. A bridge needs a detectable galvanometer current near balance; with megohm arms the currents fall to nanoamperes, swamped by leakage and thermal effects, and ratio arms of that size are impractical. The deflection method asks only for a readable deflection from a sensitive galvanometer — it is built for tiny currents.
2. I = 1.5 × 10^{-6} × 20 = 3.0 × 10^{-5} A. X = 4/(3.0 × 10^{-5}) − 80 = 133,333 − 80 ≈ 1.33 × 10^5 Ω.
3. The leakage is in parallel: effective resistance = (1.2 × 10^5 × 2 × 10^6)/(2.12 × 10^6) ≈ 1.13 × 10^5 Ω — about 6 % low. Leakage always drags the reported value down.
4. X is found by subtracting G from a total of the same order as X itself. In Experiment 1, G was the unknown and comparable to every other resistance in the circuit; here G (10^2) is a rounding term beside X (10^5).
5. Raise E (more cells — a higher voltage drives a readable current through a bigger resistance) and use a still more sensitive galvanometer (smaller k), with rigorous insulation — guard rings and PTFE insulation in serious instruments; the same formula carries through.